Online Order Estimation for Binary-valued FIR System With Colored Noise ⋆

Wenbin Wang * **Jian Guo** ** **Yanlong Zhao** *

* *State Key Laboratory of Mathematical Sciences, Academy of Mathematics and Systems Science, Chinese Academy of Sciences, Beijing 100190, China; School of Mathematical Sciences, University of Chinese Academy of Sciences, Beijing 100049, China.(e-mail: wangwenbin@amss.ac.cn, ylzhao@amss.ac.cn).*
** *Department of Applied Mathematics, The Hong Kong Polytechnic University, HongKong, China (e-mail: j.guo@amss.ac.cn)*

Abstract: This paper studies online order estimation for binary-valued finite impulse response (FIR) systems with unknown order driven by colored moving-average (MA) noise. For colored noise, the main difficulty is that temporal dependence creates long-range correlations in the binary output, which obscure the contribution of the FIR dynamics. The proposed method overcomes this by exploiting a structural property of FIR-MA models: the autocorrelation function of the underlying linear process has finite support, and this support length is preserved under binary quantization. We use this property to construct a discontinuous objective function in the candidate order, built from binary correlation statistics and designed to jump at the true support length. This objective can be evaluated recursively using only low-dimensional summary variables, without storing the full data history, and is therefore suitable for real-time implementation in the presence of colored noise. We prove that the resulting order estimator converges almost surely to the true order . In the Gaussian noise case, we further derive an explicit linear relation, which enables joint online estimation of the system order and the FIR coefficients. Numerical experiments under various noise distributions and input designs confirm the robustness and accuracy of the proposed method.

Keywords: FIR system, MA, binary-valued, order estimation, parameter identification, strong consistency.

1. INTRODUCTION

With the rapid development of information technology, the Internet of Things, and biosensing technologies, many practical systems can provide only extremely sparse, set-valued measurements-sometimes as little as a 1-bit threshold decision, i.e., a binary indicator of whether a variable exceeds a given threshold (Zhang et al., 2003). Examples include oxygen sensors in automotive exhaust and hydrocarbon sensors (Sun et al., 2004), the excitation/inhibition state of neuronal discharges (Ghysen, 2003), and low-bandwidth reporting in large sensor networks (Dargie and Poellabauer, 2010; Huang et al., 2017; Tei et al., 2014). Although 1-bit quantization is simple to implement, it severely reduces information and introduces strong nonlinearity, rendering traditional techniques such as least squares and Kalman filtering ineffective. Consequently, specialized identification methods for quantized observations have been developed.

Over the past few decades, significant theoretical progress and practical achievements have been made in the identification of systems with binary observations. Wang et al. (2003) proposed an empirical measure method to analyze the identification error, time complexity, and the impact of perturbations on system identification accuracy for linear systems under binary quantization. This method provides an effective framework for system identification under binary observation conditions, but it assumes that the noise distribution function is uniformly bounded. Building on this, Wang et al. (2018) extended the empirical measure method to finite impulse response (FIR) systems and demonstrated its asymptotic efficiency in the sense of the Cramér-Rao (CR) lower bound. Faced with the more complex autoregressive moving average (ARMA) model, Li et al. (2024) further explored how to address the strong coupling between system output and parameters when only a binary quantizer and a time-invariant system are available.

As research has progressed, many new methods have been proposed to address the computational challenges brought by binary quantization. For example, Casini et al. (2007) studied the worst-case identification of systems using binary sensors and proposed an upper bound on the time complexity for FIR systems. Zhao et al. (2016) introduced an expectation-maximization (EM)-type algorithm, proved its convergence for FIR systems, and demonstrated its exponential convergence rate, providing a robust frame-

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work for the identification of quantized systems. Similarly, You (2015) proposed a recursive estimation method based on the minimum mean square error criterion, which can adaptively adjust the quantization threshold by using set-valued innovations, thereby improving identification accuracy.

Recent research has further advanced the application of these methods in more complex systems and significantly improved the convergence rates. Guo and Zhao (2013) proposed a recursive projection algorithm for FIR systems with fixed quantization thresholds and proved its convergence at a rate of $O(\log k/k)$. Building on this, Wang et al. (2019) extended the algorithm to handle systems with matrix inputs and vector outputs, achieving a faster convergence rate of $O(1/k)$. These advancements have greatly enhanced the applicability of binary quantization identification methods in practical systems, especially in scenarios with limited bandwidth and sparse observations, providing more efficient solutions for real-world applications.

However, previous research on 1-bit quantized systems, whether for FIR or ARMA systems, has been based on the assumption that the system noise is white noise and that the system order is already known. Research on order estimation for systems with colored noise and 1-bit quantization is relatively scarce. It is known that recursive least squares yields biased estimates when the noise is colored (Söderström, 1974). While various methods have been proposed to mitigate this bias, the problem is exacerbated by binary quantization.

In fact, measurement noise is often not independent and identically distributed (i.i.d.) white noise, but rather exhibits temporal correlation (colored noise). A common and practical system is the moving average MA(q) structure, which is widely applied in systems with lag effects, such as electric load forecasting, economic variable analysis, demand prediction, and traffic flow. These systems are not only affected by current disturbances but also influenced by past factors. In 1-bit quantized systems with colored noise of the MA(q) structure, the aforementioned algorithms fail.

In statistical modeling, order estimation is a crucial step when building an accurate model, especially for FIR and MA systems. The problem of order estimation is widespread in various fields such as radar systems, power systems, seismic analysis, and physiological systems. Akaike (1974), Al-Smadi and Wilkes (1996), Chen and Guo (2012), Fuchs (1988) have proposed various methods for order estimation of ARMA models. In these papers, they derive the estimated order by minimizing a certain information criterion (such as AIC(Akaike, 1974), BIC(Weakliem, 1999), or CIC(Guo et al., 1989)). However, the estimates obtained in this way are usually non-recursive. After receiving new observations, the information criterion must be recalculated using the entire dataset, and then an optimization problem must be solved again. This process is computationally and storage-intensive.

Liang et al. (2002) leverages the relationship between the smallest eigenvalue of the covariance matrix and the Minimum Description Length (MDL) criterion to estimate the order (p, r) of an ARMA process without using coefficient estimates. Numerical examples show that the method is computationally convenient. This method later was generalized to deal with the order selection for ARMA systems with noisy observations on outputs Al-Smadi and Wilkes (1996) and for non-Gaussian processes by using the third order cumulants Al-Smadi and Wilkes (2002). Chen and Zhao (2010) provides an easily updated algorithm for estimating the order of multivariate ARMA and ARMAX processes and proves the strong consistency of the estimates.

Although research on order estimation for FIR and moving average (MA) systems is already quite extensive, these studies are all based on the assumption that the observed data are the original data. However, research on order estimation for systems with binary-valued observations is very limited.

In this paper we consider order estimation for FIR and MA systems when only binary observations are available and the measurement noise is allowed to be colored. For general noise distributions we propose online algorithms to estimate the MA order (q) and the FIR order (p). Under Gaussian noise, we further exploit the normal cumulative distribution function(CDF)/probability density function(PDF) structure to obtain linear relations between input-output cross-moments and the FIR coefficients, which permits simultaneous online estimation of model order and parameters.

The main contributions are summarized as follows:

- This paper proposes an online order estimation algorithm for binary-valued finite impulse response systems with moving-average colored noise (FIR(p)–MA(q)). The key idea is to exploit the finite-support structure of the output autocorrelation to construct a discontinuous objective function of the candidate order. Based on recursively estimated binary autocorrelations, this objective exhibits a jump at the true order. In contrast, existing methods are built on *continuous* criteria (likelihood, prediction error, or information criteria such as AIC/BIC/MDL), which typically require offline global optimization over the order, and whose flat minima can be sensitive to noise under colored disturbances. The proposed discontinuous objective produces a sharp change at the true order and can be evaluated online from low-dimensional summary statistics without storing the full data record. We prove that the selected order converges almost surely to the true one, while the algorithm greatly reduces memory and computational cost, making it suitable for binary settings.
- Under Gaussian MA noise, we exploit the CDF/PDF structure of the standard normal distribution to derive an explicit linear relation between the input-output cross-moment and the FIR coefficients. This leads to a recursive parameter estimator whose error converges at rate $O\big(\sqrt{\log\log k/k}\big)$. Together with the discontinuous order criterion, this yields a unified online procedure that identifies both the system order and the FIR coefficients under colored noise, going beyond existing binary methods that mainly address parameter estimation under white noise or known

order assumptions. Simulations under diverse noise distributions and input designs show stable order and parameter recovery, and in particular correct identification of MA lag structure even for non-Gaussian noise.

The remainder of this paper is organized as follows. Section 2 studies the binary-valued MA(q) system and its order estimator. Section 3 extends the method to FIR(p)-MA(q) systems and discusses parameter estimation under Gaussian noise. Section 4 presents simulations, and Section 5 concludes the paper.

2. BINARY-VALUED MA(q) SYSTEM AND ORDER ESTIMATION ALGORITHM

In this section, we introduce the binary-valued MA(q) system under study and present an online algorithm for estimating its order from binary observations.

2.1 System and problem setup

We consider a binary-valued MA(q) process defined as:

$$\begin{cases} y_{k+1} = \sum_{i=1}^{q} C_i \omega_{k+1-i} + \omega_{k+1} \\ s_{k+1} = \mathbb{I}_{\{y_{k+1} \geq C\}} \end{cases}, \tag{1}$$

here $y_k \in \mathbb{R}$ denotes the unobserved system output and $s_k \in \{0, 1\}$ is the binary observation obtained by the comparison between the system output and a given sensor threshold $C \in \mathbb{R}$. The coefficients $C_i \in \mathbb{R}$ are the MA parameters with $C_q \neq 0$, and $\{\omega_k\}_{k\geq 1}$ is an i.i.d. noise sequence, follows a continuous distribution with a strictly positive PDF on its support.

Problem statement and notation: Our objective is to estimate the order q from the binary observation sequence $\{s_k\}$ in an online fashion. We require an algorithm that updates a finite set of recursive statistics per time step (without storing the full history) and produces a sequence of order estimates q_k that converges almost surely to the true order. Throughout the paper, $\mathbb{E}[\cdot]$ denotes expectation and $\mathbb{P}(\cdot)$ denotes probability. We use standard shorthand for covariance and variance when needed. To guarantee the identifiability of the model order q, we assume that the binary observation process is non-degenerate (i.e., $0 < \mathbb{P}(s_k = 1) < 1$).

2.2 Estimation algorithm and theoretical results

We first present several preliminary lemmas that will be used in the analysis.

Lemma 1. Let X be a random variable with density p. Let F and G be continuous nondecreasing functions such that

$$\mathbb{E}|F(X)|,\ \mathbb{E}|G(X)|,\ \mathbb{E}|F(X)G(X)| < \infty.$$

If $\mathbb{E}[G(X)] = 0$, then $\mathbb{E}[F(X)G(X)] \geq 0$. Equivalently,

$$\int_{\mathbb{R}} F(s)G(s)p(s)\mathrm{d}s \geq 0.$$

Moreover, equality holds if F is constant on supp(p), or $G = 0$ on supp(p).

Proof. Let Y be an independent copy of X. Since F and G are nondecreasing,

$$\big(F(X) - F(Y)\big)\big(G(X) - G(Y)\big) \geq 0$$

almost surely. Hence

$$0 \leq \mathbb{E}\big[\big(F(X) - F(Y)\big)\big(G(X) - G(Y)\big)\big].$$

By independence and identical distribution,

$$\begin{aligned} &\mathbb{E}\big[\big(F(X) - F(Y)\big)\big(G(X) - G(Y)\big)\big] \\ &= 2\Big(\mathbb{E}[F(X)G(X)] - \mathbb{E}[F(X)]\mathbb{E}[G(X)]\Big). \end{aligned}$$

Since $\mathbb{E}[G(X)] = 0$, we obtain

$$\mathbb{E}[F(X)G(X)] \geq 0.$$

The equality case follows from

$$\big(F(X) - F(Y)\big)\big(G(X) - G(Y)\big) = 0 \quad a.s.$$

Thus equality holds when F is constant on supp(p), or $G = 0$ on supp(p). □

Theorem 2. Consider the binary-valued MA(q) system. If $C_q \neq 0$, then

$$\mathbb{E}[s_{k+q}s_k] \neq \mathbb{E}[s_{k+q}]\mathbb{E}[s_k].$$

Proof. Let $X = \omega_k$, and define

$$f(x) = \mathbb{P}(y_k \geq C \mid X = x), \qquad g(x) = \mathbb{P}(y_{k+q} \geq C \mid X = x).$$

Since y_k contains X with coefficient 1, f is nondecreasing. Moreover, y_{k+q} contains X with coefficient C_q. Hence g is nondecreasing if $C_q > 0$, and nonincreasing if $C_q < 0$.

Conditional on X, the remaining noise terms in y_k and y_{k+q} are independent. Therefore,

$$\mathbb{E}[s_k s_{k+q}] = \mathbb{E}[f(X)g(X)],$$

and

$$\mathbb{E}[s_k] = \mathbb{E}[f(X)], \qquad \mathbb{E}[s_{k+q}] = \mathbb{E}[g(X)].$$

Thus,

$$\begin{aligned} &\mathbb{E}[s_k s_{k+q}] - \mathbb{E}[s_k]\mathbb{E}[s_{k+q}] \\ &= \mathbb{E}\left[f(X)\big(g(X) - \mathbb{E}[g(X)]\big)\right]. \end{aligned}$$

If $C_q > 0$, then $g(x) - \mathbb{E}[g(X)]$ is nondecreasing and has mean zero. By Lemma 1,

$$\mathbb{E}[s_k s_{k+q}] - \mathbb{E}[s_k]\mathbb{E}[s_{k+q}] \geq 0.$$

If $C_q < 0$, then $-g(x) + \mathbb{E}[g(X)]$ is nondecreasing and has mean zero. Applying Lemma 1 again gives

$$\mathbb{E}[s_k s_{k+q}] - \mathbb{E}[s_k]\mathbb{E}[s_{k+q}] \leq 0.$$

Finally, by the equality condition in Lemma 1, equality can occur only if f is constant almost surely, or g is constant almost surely. Under $0 < \mathbb{P}(s_k = 1) < 1$ and $C_q \neq 0$, this degenerate case is excluded. Hence the above inequalities are strict, and therefore

$$\mathbb{E}[s_{k+q}s_k] \neq \mathbb{E}[s_{k+q}]\mathbb{E}[s_k].$$

The proof is complete. □

Remark 3. This statement formalizes the intuition that a nonzero MA parameter induces dependence between suitably lagged thresholded observations, which is the foundation for the order-identification strategy used later.

Lemma 4. (Durrett (2019), Law of the Iterated Logarithm, LIL) If $\{X_n\}$ is a sequence of i.i.d. random variables with $\mathbb{E}[X_n] = 0$ and $\mathrm{Var}(X_n) = \sigma^2 < +\infty$, then

$$\mathbb{P}\left(\limsup_{N\to\infty} \frac{\sum_{k=1}^{N} X_k}{\sqrt{2N \log\log N}} = \sigma\right) = 1. \tag{2}$$

Theorem 5. The binary process $\{s_k\}_{k\geq 1}$ is ergodic. Consequently, empirical averages and empirical autocovariances converge almost surely to their population counterparts at the rate $O\big(\sqrt{(\log\log N)/N}\big)$, i.e.

$$\frac{1}{N}\sum_{k=1}^{N} s_k - \mathbb{E}[s_k] = O\big(\sqrt{\tfrac{\log\log N}{N}}\big), \text{a.s.} \tag{3}$$

and

$$\frac{1}{N}\sum_{k=1}^{N} [s_k s_{k+i}] - \mathbb{E}[s_k s_{k+i}] = O\big(\sqrt{\tfrac{\log\log N}{N}}\big), \text{a.s.} \tag{4}$$

Remark 6. The stated rates follow from the Law of the Iterated Logarithm applied to suitably decimated subsequences; they quantify the concentration speed of the recursive estimates used by the online algorithm.

Proof. According to LIL, we have that

$$\frac{1}{N}\sum_{k=1}^{N}\big(s_{k(q+1)+i} - \mathbb{E}[s_{k(q+1)+i}]\big) = O(\sqrt{\frac{\log\log N}{N}}), \text{a.s.}$$

Then, we have that

$$\begin{aligned}\frac{1}{N}\sum_{k=1}^{N}(s_k - \mathbb{E}[s_k]) =& \frac{1}{N}\sum_{i=0}^{q}\sum_{j=0}^{\lfloor\frac{N-i}{q+1}\rfloor}\big(s_{j(q+1)+i} - \mathbb{E}[s_{j(q+1)+i}]\big)\\ =& O(\sqrt{\frac{\log\log N}{N}}), \text{a.s.}\end{aligned}$$

Then we prove the first part of the theorem. Similarly, we can prove the second part of the theorem, and the details are omitted here for brevity. □

We now present a simple online algorithm for estimating the order q of the binary-valued MA(q) process.

- $g_k = (1-\frac{1}{k})g_{k-1} + \frac{1}{k}s_k$
- $h_k(j) = (1-\frac{1}{k})h_{k-1}(j) + \frac{1}{k}s_k s_{k+j}, j = 1, 2, \cdots, M$
- $\gamma_{j,k} = h_k(j) - g_k^2, j = 1, 2, \cdots, M$

We define the following population autocovariances:

$$\gamma_j = \mathbb{E}[s_k s_{k+j}] - \mathbb{E}[s_k]\mathbb{E}[s_{k+j}] \tag{5}$$

Then we have the following corollary.

Corollary 7. It holds that

$$\lim_{k\to\infty}\gamma_{j,k} = \gamma_j, \text{a.s.} \tag{6}$$

and the rate of convergence is given by

$$\gamma_{j,k} - \gamma_j = O(\sqrt{\frac{\log\log k}{k}}), \text{a.s.} \tag{7}$$

Define the following decision index:

$$Q_{j,k}^{(q)} \triangleq \frac{|\gamma_{j+1,k}| + r(k)}{|\gamma_{j,k}| + r(k)}$$

where $r(k)$ is a regularization term satisfying that

$$r(k) > 0, \lim_{k\to\infty} r(k) = 0$$

and

$$\lim_{k\to\infty}\frac{\sqrt{\frac{\log\log k}{k}}}{r(k)} = 0.$$

Remark 8. The regularization term $r(k)$ is introduced to avoid numerical instability when $|\gamma_{j,k}|$ is very small. The conditions on $r(k)$ ensure that it vanishes asymptotically but at a slower rate than the estimation error of $\gamma_{j,k}$, which is crucial for the consistency proof. Usually, a choice like $r(k) = \frac{1}{k^\alpha}$ with $\alpha \in (0, 0.5)$ suffices. Another possible choice is $r(k) = \frac{1}{\log k}$.

Then take a threshold $\varepsilon \in (0,1)$, say $\varepsilon = 0.5$, and estimate q as:

$$q_k = \arg\max\{j : Q_{j,k}^{(q)} \leq \varepsilon, j = 1, 2, \cdots, M\}.$$

Then we have the following main theorem to guarantee the consistency of the estimator q_k.

Theorem 9. For the binary-valued MA(q) process defined above, then the estimator q_k converges to the true order q almost surely, i.e.,

$$\mathbb{P}\left(\lim_{k\to\infty} q_k = q\right) = 1.$$

Proof. According to Corollary 7, we have that

$$\lim_{k\to\infty} Q_{j,k}^{(q)} = \begin{cases}\frac{|\gamma_{j+1}|}{|\gamma_j|}, & \gamma_j \neq 0\\ 1, & \gamma_j = 0, \gamma_{j+1} = 0\\ \infty, & \gamma_j = 0, \gamma_{j+1} \neq 0\end{cases}$$

We notice that $\gamma_j = 0, \forall j > q$ and $\gamma_q \neq 0$ according to Theorem 2 . Thus, it holds that

$$\lim_{k\to\infty} Q_{j,k}^{(q)} = \begin{cases}0, & j = q\\ 1, & j > q\end{cases}$$

which implies that

$$\lim_{k\to\infty} q_k = q,$$

which completes the proof. □

3. BINARY-VALUED FIR(p)-MA(q) SYSTEM AND ORDER ESTIMATION ALGORITHM

In this section, we extend the previous results to the binary-valued FIR(p)-MA(q) system and present an online algorithm for estimating its order p and the parameter.

3.1 System and problem setup

Consider the following binary-valued FIR(p)-MA(q) stochastic system:

$$\begin{cases} y_{k+1} = \sum_{i=1}^{p}\theta_i u_{k+1-i} + \sum_{i=1}^{q} C_i\omega_{k+1-i} + \omega_{k+1}\\ s_{k+1} = \mathbb{I}_{\{y_{k+1}\geq C\}}\end{cases} \tag{8}$$

where $u_k \in \mathbb{R}$ is the input and $y_k \in \mathbb{R}$ is the unobserved system output. The binary observation $s_k \in \{0,1\}$ is obtained by thresholding y_k at $C \in \mathbb{R}$. The parameters θ_i and C_i are real valued with $\theta_p \neq 0$ and $C_q \neq 0$. The noise sequence $\{\omega_k\}_{k\geq 1}$ is i.i.d.

Problem statement and notation: Our objective is to estimate the binary-valued FIR order p from the binary observation sequence $\{s_k\}$ and the input sequence $\{u_k\}$ in an online fashion. We require an algorithm that updates a finite set of recursive statistics per time step (without storing the full history) and produces a sequence of order estimates p_k that converges almost surely to the true order.

3.2 Estimation algorithm and theoretical results

We first present a preliminary lemma that will be used in the analysis.

Lemma 10. For the binary-valued FIR(p)-MA(q) system, assume that $\{u_k\}$ is i.i.d., independent of $\{\omega_k\}$, and satisfies $\mathbb{E}u_k = 0$. If $\theta_p \neq 0$ and the input is non-degenerate, then

$$\mathbb{E}[u_k s_{k+p}] \neq 0.$$

Proof. Let $U = u_k$. Since y_{k+p} contains u_k only through the term $\theta_p u_k$, we can write

$$y_{k+p} = \theta_p U + \xi_k,$$

where ξ_k is independent of U. Define

$$\phi(u) = \mathbb{P}(\theta_p u + \xi_k \geq C).$$

Then

$$\mathbb{E}[u_k s_{k+p}] = \mathbb{E}\big[U\mathbb{P}(y_{k+p} \geq C \mid U)\big] = \mathbb{E}[U\phi(U)].$$

If $\theta_p > 0$, then ϕ is nondecreasing. Since $\mathbb{E}U = 0$, Lemma 1 gives

$$\mathbb{E}[U\phi(U)] \geq 0.$$

If $\theta_p < 0$, then $-\phi$ is nondecreasing, and Lemma 1 gives

$$\mathbb{E}[U\phi(U)] \leq 0.$$

By Lemma 1, equality can occur only if $U = 0$ a.s. or ϕ is constant on the support of U. Under the non-degeneracy assumption, this is excluded. Hence

$$\mathbb{E}[u_k s_{k+p}] \neq 0.$$

□

Remark 11. This lemma shows that when the last FIR parameter θ_p is nonzero, then the lag-p cross moment between the input and the binary output is nontrivial, which motivates the order estimation algorithm presented next.

We now present a simple online algorithm for estimating the order of the binary-valued FIR(p)-MA(q) systems.

- $\beta_{j,k} = (1 - \frac{1}{k})\beta_{j,k-1} + \frac{1}{k}u_k s_{k+j}$, for $j = 0, 1, \ldots, M$.

We define:

$$\beta_j = \mathbb{E}[u_k s_{k+j}] \tag{9}$$

Corollary 12. It holds that, for $j = 0, 1, \ldots, M$,

$$\lim_{k\to\infty} \beta_{j,k} = \beta_j, \text{a.s.} \tag{10}$$

and

$$\beta_{j,k} - \beta_j = O(\sqrt{\frac{\log\log k}{k}}), \text{a.s.} \tag{11}$$

Define the following decision index:

$$Q_{j,k}^{(p)} \triangleq \frac{|\beta_{j+1,k}| + r(k)}{|\beta_{j,k}| + r(k)}$$

Then take a threshold $\varepsilon \in (0, 1)$, say $\varepsilon = 0.5$, and estimate p as:

$$p_k = \arg\max\{j : Q_{j,k}^{(p)} \leq \varepsilon,\ j = 1, 2, \ldots, M\}.$$

Theorem 13. For the binary-valued FIR(p)-MA(q) system defined above, if $\theta_p \neq 0$, then the estimator p_k converges to the true order p almost surely, i.e.,

$$\mathbb{P}\left(\lim_{k\to\infty} p_k = p\right) = 1.$$

3.3 Gaussian noise condition

In this subsection, we consider the case where $\{\omega_k\}_{k\geq 1} \sim \mathcal{N}(0, \sigma^2)$. In this case, we can further estimate the FIR coefficients θ_i while estimating the order p.

To streamline the presentation, we treat the Gaussian-noise case by splitting it into two subcases. First, when the threshold $C = 0$ there is a scale indeterminacy: multiplying all FIR coefficients by a common factor leaves the binary observations unchanged, so parameter recovery requires knowledge of (or a consistent estimator for) the output variance R. Second, when $C \neq 0$ the Gaussian assumption yields explicit relations (involving the standard normal CDF Φ and PDF ϕ) between input-output cross-moments and the unknown FIR coefficients; these relations allow simultaneous estimation of the model order and parameters with provable consistency and convergence rates. We analyze each subcase below and present the corresponding lemmas, algorithms and theoretical guarantees.

Threshold C=0 When the threshold $C = 0$, the system is not identifiable under scale transformations: multiplying all FIR coefficients by a common factor leaves the distribution of the binary observations $\{s_k\}$ unchanged. Thus, parameter recovery requires knowledge of (or a consistent estimator for) the output variance $R = \mathrm{Var}(y_k)$.

Lemma 14. For the binary-valued FIR(p)-MA(q) system defined above with threshold $C = 0$, it holds that

$$\mathbb{E}[u_k s_{k+i}] = \frac{R_{y,u}(i)}{\sqrt{2\pi R}}, \tag{12}$$

where $R = \mathrm{Var}(y_k)$ and $R_{y,u}(i) = \mathrm{Cov}(y_{k+i}, u_k)$.

Thus, we can estimate

- $\beta_{j,k} = (1 - \frac{1}{k})\beta_{j,k-1} + \frac{1}{k}u_k s_{k+j}$, for $j = 0, 1, \ldots, p_k$.
- $\theta_{j,k} = \sqrt{2\pi R}\,\beta_{j,k}$.

Then we have the following theorem to guarantee the convergence of the parameter estimator $\theta_{j,k}$.

Theorem 15. For the binary-valued FIR(p)-MA(q) system with threshold $C = 0$, assume $\omega_k \sim \mathcal{N}(0, \sigma^2)$ (with unknown σ^2) and the stochastic input $u_t \sim \mathcal{N}(0, 1)$ is independent of $\{\omega_k\}$. Then the parameter estimator $\theta_{j,k}, j = 1, 2, \cdots, M$ satisfies

$$\theta_{j,k} - \theta_j = O\Big(\sqrt{\frac{\log\log k}{k}}\Big), \quad \text{a.s.} \tag{13}$$

Threshold $C \neq 0$ When the threshold $C \neq 0$, we can estimate the parameters by the binary observations $\{s_k\}$ directly.

Lemma 16. For the binary-valued FIR(p)-MA(q) system with Gaussian noise, assume that $\{u_k\}$ is i.i.d. with $\mathbb{E}u_k = 0, \mathbb{E}u_k^2 = 1$, and independent of $\{\omega_k\}$. Let $R = \mathrm{Var}(y_k)$ and $\alpha = C/\sqrt{R}$. Then

$$\mathbb{E}[s_k] = 1 - \Phi(\alpha) = \Phi(-\alpha),$$

and, for $i \geq 1$,

$$\mathbb{E}[u_k s_{k+i}] = \begin{cases} \dfrac{\phi(\alpha)}{\sqrt{R}}\theta_i, & 1 \leq i \leq p, \\ 0, & i > p. \end{cases}$$

where Φ and ϕ denote the CDF and PDF of the standard normal distribution, respectively.

Remark 17. This lemma shows that when the noise is Gaussian, then there exists a linear relation between the cross moment $\mathbb{E}[u_k s_{k+i}]$ and the FIR coefficient θ_i, which allows us to estimate the FIR coefficients once we have estimated the order p.

Thus, we can estimate

- $g_k = (1 - \frac{1}{k})g_{k-1} + \frac{1}{k}s_k$
- $\beta_{j,k} = (1 - \frac{1}{k})\beta_{j,k-1} + \frac{1}{k}u_k s_{k+j}, j = 0, 1, 2, \cdots, p_k$
- $\alpha_k = -\Phi^{-1}(g_k)$
- $\theta_{j,k} = \frac{C}{\alpha_k \phi(\alpha_k)}\beta_{j,k}$

Then we have the following theorem to guarantee the convergence of the estimator $\theta_{j,k}$.

Theorem 18. For the binary-valued FIR(p)-MA(q) system defined above, if $\omega_k \sim \mathcal{N}(0, \sigma^2)$, the variance σ^2 is unknown, we stochastic input $u_t \sim \mathcal{N}(0, 1)$, and u_t is independent with ω_k, then the estimator $\theta_{j,k}$, for $j = 1, \ldots, M$ converges to the true parameters almost surely, i.e.,

$$\theta_{j,k} - \theta_j = O\left(\sqrt{\frac{\log\log k}{k}}\right), \text{a.s.} \tag{14}$$

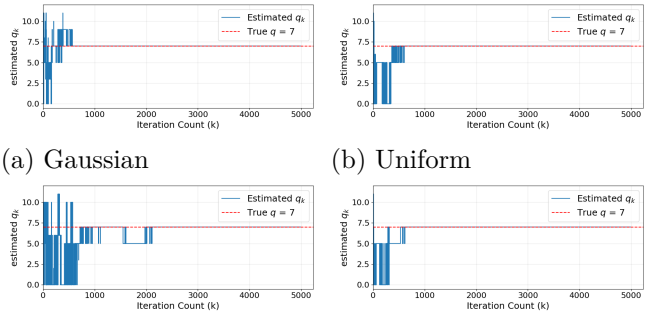
4. SIMULATION RESULTS

In this section, we present some simulation results to illustrate the performance of the proposed online order estimation algorithms for both binary-valued MA(q) and binary-valued FIR(p)-MA(q) systems.

Example 19. We first test the proposed MA-order estimator on the binary-valued MA system in (1). The true order is $q = 7$, the threshold is $C = 0$, and the MA coefficients are

$$(C_1, \ldots, C_7) = (0.8, -0.5, 0.3, -0.2, -0.9, -0.3, 1.0).$$

We set $M = 12$ and $r(k) = k^{-0.3}$. Four noise distributions are considered: Gaussian, uniform, exponential, and Laplace. The corresponding results are shown in Fig. 1. The estimator converges to the true order under all four noise distributions, illustrating the robustness of the proposed method to non-Gaussian colored noise.



(a) Gaussian (b) Uniform

(c) Exponential (d) Laplace

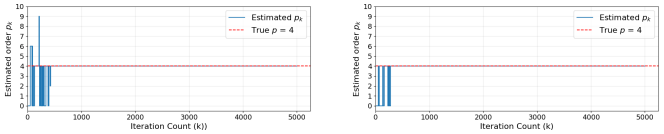
Fig. 1. Order estimates of the binary-valued MA(7) system under different noise distributions.

From Figs. 1, we can see that the proposed online order estimation algorithm can correctly estimate the order of the MA(7) process under different noise distributions.

Example 20. We next consider the binary-valued FIR(4)-MA(3) system in (8), with

$$\theta = (0.8, -0.5, 0.3, -1.0), \qquad (C_1, C_2, C_3) = (0.3, -0.4, 0.2).$$

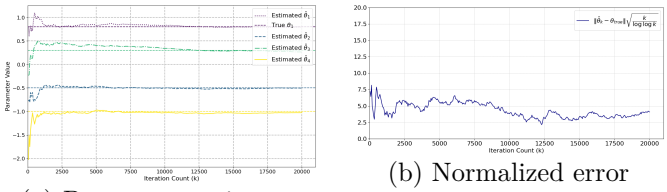
The noise is $\omega_k \sim \mathcal{N}(0, 1)$, the threshold is $C = 0$, and we set $M = 10$ and $r(k) = k^{-0.3}$. Two stochastic inputs are tested: $u_k \sim \mathcal{N}(0, 1)$ and $u_k \sim U(-1, 1)$. As shown in Fig. 2, the proposed estimator correctly recovers the FIR order under both input distributions.



(a) Gaussian input (b) Uniform input

Fig. 2. Order estimates of the binary-valued FIR(4)-MA(3) system under different stochastic inputs.

Finally, for the same FIR-MA system with Gaussian input, we set $C = 0.5$ and estimate the FIR coefficients using the Gaussian-noise relation in Lemma 16. The parameter estimates and the normalized error trajectory are reported in Fig. 3. The results show accurate parameter recovery, and the normalized error is consistent with the theoretical rate $O(\sqrt{(\log\log k)/k})$.



(a) Parameter estimates (b) Normalized error trajectory

Fig. 3. Parameter estimation performance for the binary-valued FIR(4)-MA(3) system with Gaussian input and threshold $C = 0.5$.

5. CONCLUSION

This paper has addressed online order estimation for binary-valued FIR(p)-MA(q) systems under colored noise. Exploiting the finite-support structure of the output autocorrelation, we constructed a discontinuous objective function of the candidate order, built from recursively updated binary correlation statistics. This objective exhibits a sharp change at the true order and can be evaluated from low-dimensional summary variables without storing the full data record. From a theoretical viewpoint, we showed that the resulting order estimator is almost surely consistent and established its convergence rate. In the Gaussian MA case, we further derived an explicit linear relation between the input-output cross-moment and the FIR coefficients, and on this basis proposed a recursive parameter estimator with provable convergence rate. Taken together, these results yield a unified online method that performs joint order and parameter identification under colored noise, going beyond existing binary identification approaches that typically assume white noise or a known model order. Simulations under a range of input designs and noise distributions confirmed stable order selection and accurate parameter recover.

As a direction for future work, we plan to extend the framework to include autoregressive components, so as to handle more general ARMA dynamics within the same binary and colored-noise setting.

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